

SALMAN IBNEY RAHMAN

Full-Stack Developer & Data Scientist

✉ rahmanmdsalman428@gmail.com ☎ +8801886455815

📍 Dhaka, Bangladesh 🔗 [linkedin.com/in/salman-ibney-rahman/](https://www.linkedin.com/in/salman-ibney-rahman/)



SUMMARY

Recent Computer Science graduate with hands-on experience building full-stack web platforms and machine learning pipelines. Proficient in Next.js, Express, MongoDB, Python, Scikit-learn, and TensorFlow with production deployments on Vercel and cloud platforms. Delivered complete systems including JWT authentication, real-time features, data processing pipelines, and ML model integration. Strong Git workflows, CI/CD automation, and technical documentation skills. Seeking software engineering or data science roles requiring quick learning and end-to-end execution.

TECHNICAL & DIGITAL SKILLS

ML / Data Science

Scikit-learn, TensorFlow, Keras, NLTK, Pandas, NumPy, Matplotlib, Seaborn

Tools & Platforms

Git, GitHub, Jupyter Notebook, VS Code, Arduino IDE, Processing IDE, LaTeX, Kaggle, Google Colab

Backend & Database

Node.js, Express.js, MongoDB Atlas, RESTful API Design & Integration

Cloud & DevOps

Git, GitHub (branches, PRs, merges), Vercel, Netlify, Render, CI/CD

Programming Languages

Python, C, C++, Java, JavaScript

Frontend Ecosystem

JavaScript (ES6+, DOM, Fetch), React.js (Hooks, Router), Next.js 14 (App Router), HTML5, CSS3

Authentication & Security

BetterAuth, Role-Based Access Control (RBAC), Session Management, Protected Route Middleware

PROJECTS

PetNest — Full-Stack Pet Adoption Platform [🔗](#)

Live: pet-nest-adopt.vercel.app [🔗](#) | **GitHub:** github.com/salmanibneyrahman/PH-A-09-PetNest [🔗](#) (frontend) | <https://github.com/salmanibneyrahman/PetNest-Server> [🔗](#) (backend)

Stack: Next.js 16 (App Router, SSR), React 19 (Server Components), Express.js 5, MongoDB Atlas, Better Auth (JWT/JWKS), HeroUI, Framer Motion, Tailwind, Vercel

- Shipped a production full-stack pet-adoption platform with Next.js 16 App Router and React 19 Server Components, deployed on Vercel with SSR achieving 1.2s First Contentful Paint and a 92 Lighthouse Performance score.
- Built end-to-end authentication using Better Auth with JWKS cryptographic verification, JWT stored in HTTPOnly cookies, and route-level protection with callback-URL preservation across login redirects.
- Engineered a real-time adoption-request system using silent 10-second background polling with functional setState to prevent memory leaks, surfacing pinpoint Facebook-style notifications on new requests.
- Implemented owner-authorized CRUD across pets and adoption requests, with MongoDB `\$regex` / `\$in` queries for multi-field search and a state-machine workflow (pending → approved/rejected) that auto-rejects competing requests when one is approved.
- Designed an embedded wishlist schema (array field on the user document) reducing database queries ~50%, and built a custom Framer-Motion toast system saving ~15 KB bundle size versus react-hot-toast.

SunCart — Full-Stack E-Commerce Platform [🔗](#)

Live: ph-a-08-sun-cart.vercel.app [🔗](#) | **GitHub:** github.com/salmanibneyrahman/PH-A-08-SunCart [🔗](#)

Technologies: Next.js 14, React 18, Tailwind CSS, DaisyUI, MongoDB Atlas, BetterAuth, Vercel, Render.

- Developed full-stack e-commerce application using Next.js 14 App Router and Server-Side Rendering (SSR) to reduce initial page load times and optimize SEO indexing.
- Connected external RESTful APIs with a MongoDB Atlas cluster to handle real-time product fetching, multi-criteria filtering, smart sorting, and debounced search queries.

- Implemented session-based authentication using BetterAuth, enforcing Role-Based Access Control (RBAC) across protected middleware routes and ensuring production HTTPS security.
- Designed a responsive glassmorphic UI featuring a dual grid/list catalog view toggle, continuous promo marquees, and interactive hover animations using Tailwind CSS and DaisyUI.
- Configured a MongoDB Atlas cloud cluster using the native MongoDB Node Client Driver Adapter to manage scalable and distributed data storage.
- Optimized performance metrics by utilizing dynamic code splitting, React computation memoization, Next.js Image lazy loading images with placeholder blurs, and an automated Vercel CI/CD pipeline.

Machine Learning Based Climate-Resilient Crop Recommendation and Yield Prediction for Bangladesh

Capstone Project

Technologies: Python, Classification & Regression models, Streamlit, Data Science, Predictive Modeling.

- Development: Developed an end-to-end ML pipeline to recommend climate resilient crops and predict yield based on environmental parameters.
- Deployment & DX: Deployed a user-facing Streamlit application integrating GPS auto-detection and real-time weather data via Open-Meteo API, transforming a complex ML model into an intuitive tool for end-users, significantly improving accessibility and UX.
- Documentation: Authored comprehensive report & technical guides to ensure the project is reproducible, allowing other developers to set up the environment and run the model with minimal friction.

Credit Card Fraud Detection — Financial Risk and Data Analytics

Machine Learning Project

- Engineered a fraud-detection analytical framework for severely imbalanced transaction datasets, achieving F1 = 0.92 and demonstrating quantitative methods directly applicable to banking fraud prevention and credit-risk assessment
- Applied SMOTETomek class-balancing techniques and benchmarked five analytical models (LightGBM, Autoencoder, TabNet, LSTM, Stacked Ensemble) against precision, recall, and F1 metrics for comparative risk analysis
- Produced a structured research-grade report documenting methodology, model comparison metrics, and technical findings in a format aligned with audit and compliance documentation standards

EDUCATION

Bachelor of Science in Computer Science & Engineering

2022 – 2026

East West University

Dhaka, Bangladesh

- CGPA: 3.17

LANGUAGES

Bengali

Native

English

Professional Proficiency

Hindi

Conversational

ACHIEVEMENTS

Research Leadership

East West University

- Led capstone research on ML-based climate-resilient agricultural prediction, managing the full lifecycle from data collection to deployment.

Academic Excellence

East West University

- Data analysis, financial data interpretation, predictive modeling, statistical analysis, and report generation

Board Excellence

BAF Shaheen College Chattogram / Bangladesh Bank Colony High School

- Achieved a perfect GPA of 5.00 out of 5.00 in both Higher Secondary Certificate (HSC) and Secondary School Certificate (SSC).

Project Portfolio

Self

- Successfully delivered 9+ diverse projects spanning ML, networking, and cybersecurity, all maintained with professional-grade documentation on GitHub.